

■ BOAR System Architecture & RAG Benchmark Manual

SPECIFICATION & TEST DATASET FOR OFFLINE HYBRID SEARCH (BOUNTY VERIFICATION)

1. Hardware Profile & Operational Constraints

The BOAR (Best Offline AI Researcher) platform is strictly designed to operate within severe mobile hardware limits. It achieves zero-latency off-grid intelligence without cloud dependencies or active internet connection. All model inference, vector similarity search, and full-text retrieval are executed natively on Android devices (e.g., GrapheneOS / Xiaomi HyperOS).

RAM Overhead Ceiling 12.0 GB Physical Limit	Storage Footprint Cap 50.0 GB Total Ceiling	Target Operating System Android 14+ / GrapheneOS
Native Inference Engine llama.cpp / llama.rn	Hybrid Search Database SQLite FTS5 + sqlite-vec	Vector Embedding Model all-MiniLM-L6-v2 (384-d)

2. Custom Document Ingestion & Chunking Parameters

When custom files (PDFs, Markdown `.md`, or TXT notes) are uploaded into BOAR, the on-device parser triggers an automated vector indexing pipeline. Text is chunked with precise boundaries to maximize precision during retrieval-augmented generation:

- **Chunk Window Size:** 500 tokens (~2,000 characters) per segment.
- **Overlap Window:** 50 tokens (~200 characters) between adjacent chunks to preserve contextual continuity.
- **Vector Dimension:** 384-dimensional dense float vectors generated via localized ONNX runtime.
- **Full-Text Search Index:** SQLite FTS5 table with BM25 ranking for exact keyword matching.
- **Reciprocal Rank Fusion (RRF):** Combined score equation: $Score = 1 / (60 + Rank_Vector) + 1 / (60 + Rank_BM25)$.

■ Key Test Fact for RAG Verification

Unique Verification Code: BOAR-RAG-9982-ALPHA.

Field Protocol Code Name: Project Nightfall.

Emergency Frequency Channel: 146.520 MHz VHF.

Test Question to ask your app: 'What is the unique verification code and emergency frequency channel for Project Nightfall?'

3. Model Quantization & Performance Matrix

To balance generation speed (tokens/second) against active process memory consumption (RSS), BOAR categorizes models into three hardware tiers:

Model Name	Quantization	Param Count	RAM Footprint	Tokens/Sec	Compatibility Rating
Qwen 2.5 1.5B	Q4_K_M	1.54 Billion	1.2 GB	32.4 t/s	■ Runs Great (All Phones)
Llama 3.2 3B	Q4_K_M	3.21 Billion	2.1 GB	21.8 t/s	■ Runs Great (6GB+ RAM)
Qwen 2.5 7B	Q4_K_M	7.61 Billion	4.8 GB	12.5 t/s	■ High RAM (8GB+ RAM)
Llama 3.1 8B	Q4_K_M	8.03 Billion	5.2 GB	9.6 t/s	■ High RAM (12GB RAM)
Qwen 2.5 14B	Q2_K	14.7 Billion	7.8 GB	4.2 t/s	■ Heavy (Requires 12GB)

4. RAG Retrieval Ground-Truth Benchmark Queries

Use the following ground-truth questions and expected answers to test your custom PDF indexing inside the BOAR app:

Test Query Number	User Question Prompt	Expected Ground Truth Answer in Document
RAG-TEST-01	What is the RRF formula used in BOAR?	Score = 1 / (60 + Rank_Vector) + 1 / (60 + Rank_BM25)
RAG-TEST-02	What is the token chunk size and overlap?	500 tokens chunk size with a 50 token overlap
RAG-TEST-03	What is the hardware RAM ceiling for the bounty?	12.0 GB physical RAM limit (50.0 GB storage cap)
RAG-TEST-04	Which embedding model is used for local vectors?	all-MiniLM-L6-v2 generating 384-dimensional vectors
RAG-TEST-05	What emergency frequency is specified for Nightfall?	146.520 MHz VHF under Project Nightfall

Generated automatically for BOAR Android App RAG Verification testing. Fully grounded in repository technical specs.